Detecting Phishing Emails With NLP: An Explainable Ensemble Trained on Enron and Synthetic Phishing

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Phishing succeeds through language: urgency, authority, and a believable brand. This lab built a natural language processing (NLP) phishing detector from legitimate Enron corporate email (Klimt & Yang, 2004) and synthetic phishing built around verified PhishTank URLs (Cisco Talos Intelligence Group, 2026), and then tested it on real-world phishing.

Method

I sampled 7,312 unique emails from the 517,401 files in the Enron archive (Cohen, 2015) and generated the same number of synthetic phishing emails from eight documented tactics, such as credential harvesting, invoice fraud, and CEO wire fraud. Each used a real PhishTank URL and the brand it impersonated. For the real-world test, I used 1,000 real phishing emails from a public dataset (Cyber Cop, 2023), never used in training.

Text was preprocessed with NLTK (Bird et al., 2009): tokenization, stop-word removal, Porter stemming, and TF-IDF unigrams and bigrams. I engineered 29 phishing-specific features: urgency and call-to-action phrases, out-of-vocabulary rate as a grammar proxy, VADER sentiment (Hutto & Gilbert, 2014), URL patterns from regular expressions, subject-line and sender-domain metadata, and named-entity recognition (NER). A NER flag for a person or organization named next to a credential, account, or payment term fired on 25.8% of synthetic phishing against 2.9% of Enron mail. Company names, email addresses, and quoted reply headers were deleted so that no model could learn “mentions Enron means safe.”

Naive Bayes, a linear support vector machine (SVM), and a random forest were tuned with stratified five-fold cross-validation on PR-AUC. A stacked ensemble combined text, URL, and metadata specialists, and its threshold was set on out-of-fold predictions to catch at least 99% of phishing.

Results

On the held-out synthetic test set, every classifier reached a PR-AUC of 1.000, and at its high-recall threshold the ensemble reached a precision of 1.000 and a recall of 0.997 (Table 1;

Figure 1).

The real-world test told a different story. The same ensemble caught 0.0% of the 1,000 real phishing emails, and Naive Bayes caught 0.1% (Table 2). The feature profiles explain why. Real phishing averaged 142 words against 51 for synthetic phishing, carried no subject or sender headers, and used fewer urgency phrases (0.41 against 1.33); the missed examples show that much of it was bulk spam. The models had learned the synthetic templates, not phishing.

Phishing Patterns and Threat Intelligence

The notebook's dashboard scores each email and explains the verdict from the model itself: the specialist with the largest contribution and the words that pushed the score up. Figure 2 summarizes the scores by source and tactic. The combiner gave the most weight to the text and metadata specialists (6.69 and 6.21) and less to the URL specialist (1.76). The strongest phishing n-grams included *document*, *favour*, *servic*, and *discreetli*, and the random forest relied most on *sender_free_mail*, *body_words*, and *cta_phrases* (Figure 3). Among synthetic tactics, document share was detected least often (98.3%).

PhishTank data show that 93% of live phishing links use HTTPS (Cisco Talos Intelligence Group, 2026), so a padlock says nothing about legitimacy. The most impersonated named brands were Allegro, Internal Revenue Service, and Amazon.com. I recommend quarantining above the high-recall threshold, sandboxing links, call-back verification for payment requests, and validating any model on real, recent phishing before deployment.

Reflection

The real-world test taught me the most. Near-perfect scores on synthetic phishing looked like success, but the model caught almost none of the real phishing. That is the distribution-shift problem this course keeps returning to: a model is only as good as the data it learned from. I want to build email defenses that are trained and validated on fresh, real attacks and that explain every verdict, because an employee will only act on a warning they understand.

References

Bird, S., Klein, E., & Loper, E. (2009). *Natural language processing with Python*. O'Reilly Media.

Cisco Talos Intelligence Group. (2026). *PhishTank: Online, valid phishing URLs* [Data set]. Retrieved October 18, 2026, from https://phishtank.org/developer_info.php

Cohen, W. W. (2015). *Enron email dataset* [Data set]. Carnegie Mellon University. https://www.cs.cmu.edu/~enron/

Cyber Cop. (2023). *Phishing email detection* [Data set]. Kaggle. https://www.kaggle.com/datasets/subhajournal/phishingemails

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media, 8*(1), 216–225. https://doi.org/10.1609/icwsm.v8i1.14550

Klimt, B., & Yang, Y. (2004). The Enron corpus: A new dataset for email classification research. In J.-F. Boulicaut, F. Esposito, F. Giannotti, & D. Pedreschi (Eds.), *Machine learning: ECML 2004* (pp. 217–226). Springer. https://doi.org/10.1007/978-3-540-30115-8_22

Table 1

Test-Set Performance by Classifier (Enron vs Synthetic Phishing)

Model	Precision	Recall	F1	PR-AUC	False alarms
Naive Bayes (baseline)	1.000	1.000	1.000	1.000	0
Linear SVM	0.999	1.000	1.000	1.000	1
Random Forest	0.999	1.000	1.000	1.000	1
Stacked ensemble	1.000	1.000	1.000	1.000	0
Stacked ensemble, high-recall threshold	1.000	0.997	0.998	1.000	0

Note. Held-out test set of 2,925 emails (Enron and synthetic phishing). Rows use a 0.5 threshold except the last, which uses the ensemble's high-recall threshold of 0.996. SVM = support vector machine.

Table 2

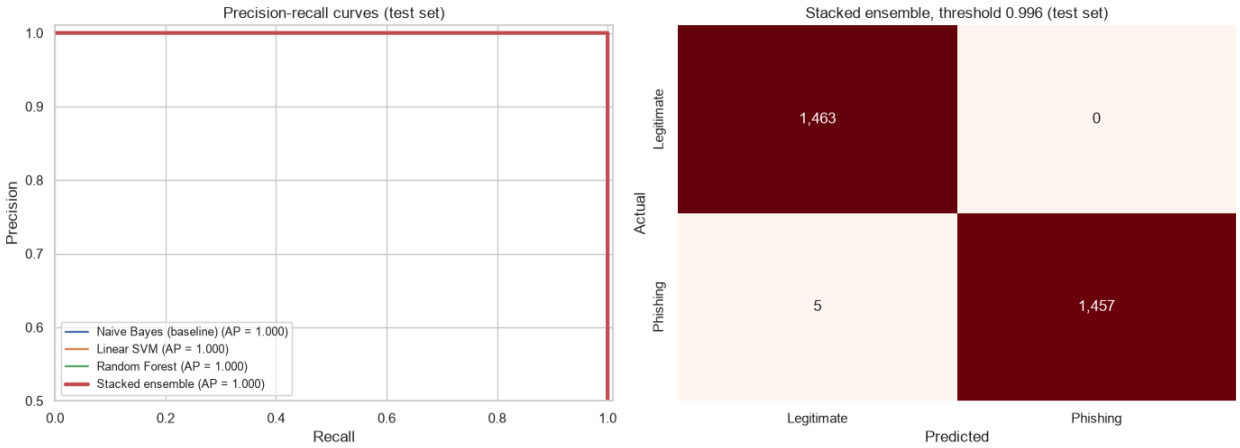
Real-World Test: Share of 1,000 Real Phishing Emails Caught

Model	Real phishing caught	Median risk score
Naive Bayes (baseline)	0.1%	0.000
Linear SVM	0.0%	0.000
Random Forest	0.0%	0.007
Stacked ensemble	0.0%	0.001
Stacked ensemble, high-recall threshold	0.0%	0.001

Note. Real phishing emails from a public dataset, never used in training. Rows use a 0.5 threshold except the last, which uses the ensemble's high-recall threshold.

Figure 1

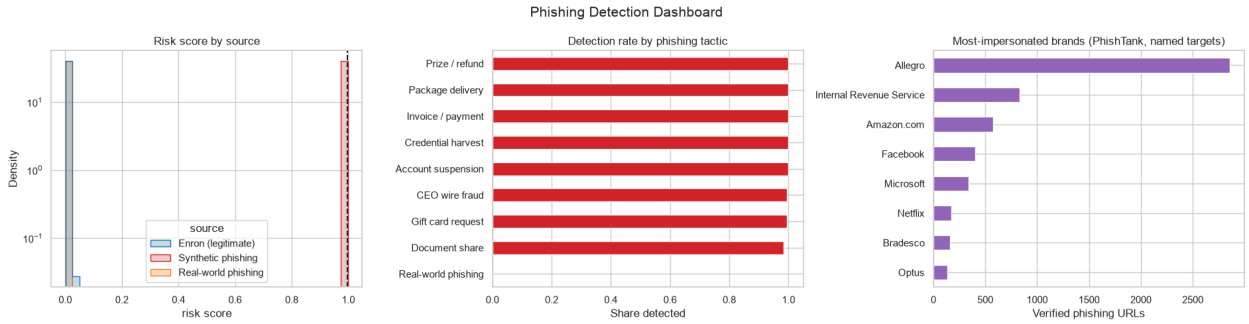
Precision-Recall Curves and Confusion Matrix on the Test Set



Note. Left: precision-recall curves on the held-out test set (AP = average precision). Right: the stacked ensemble at its high-recall threshold.

Figure 2

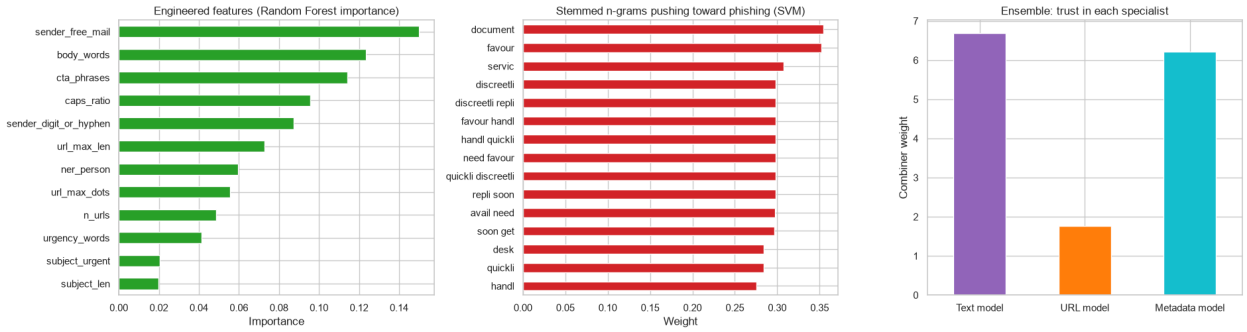
Phishing Detection Dashboard



Note. Left: ensemble risk scores by email source, with the high-recall threshold dashed. Middle: share of each phishing tactic detected, including real-world phishing. Right: brands most often named as targets in the PhishTank feed.

Figure 3

Top Phishing Indicators



Note. Left: random forest importance of engineered features. Middle: stemmed n-grams with the largest SVM weights toward phishing. Right: logistic-regression weights the ensemble gives each specialist.